

Q1:JUN 2016 P2

- 7 (a) Distinguish between *evaporation* and *boiling*.

[2]

- (b) Explain why a wet person standing in a windy place feels cold.

[3]

Q2:NOV 2013 P2

- 6 (a) The specific latent heat of fusion of a substance is $3.34 \times 10^5 \text{ J kg}^{-1}$.
Explain this statement.

[2]

- (b) Explain why the value of latent heat of fusion of a substance is less than that of latent heat of vaporisation.

[2]

6.4 THERMAL PROPERTIES OF MATERIALS P2, P3&P5 QUESTIONS

Compiled by: R. Muza

- (c) Smoke particles were used in an experiment to demonstrate the **Brownian motion**.

- (i) Describe the observations in the experiment.

- (ii) State, with a reason, the effect on the speed when large smoke particles are used.

[4]

Q3:NOV 2008 P5

- 4 (a) Define the term *specific latent heat*. [2]

- (b) Table 4.1 shows some of the thermal properties of copper and gold that you may use.

Table 4.1

	melting point/ °C	specific heat capacity/Jkg ⁻¹ K ⁻¹	specific latent heat of fusion/Jkg ⁻¹ × 10 ⁴
copper	1356	132	21.0
gold	1340	385	7.0

- (i) Power of 1.5 kW is supplied for 30 minutes to solid copper which has just reached its melting point and it just melts, determine the mass of the copper.
- (ii) By defining internal energy of a system, explain why there is an increase in internal energy during melting.
- (iii) Some wedding rings can be made from a mixture of copper and gold. The mass of copper calculated in (b)(i) has its temperature raised to 1 400°C. Gold at 500°C was added to the copper until the temperature of the mixture was 1 375°C.

Determine the mass of gold that was added. [Assume specific heat capacity for both metals is the same in the liquid and solid states]

[10]

6.4 THERMAL PROPERTIES OF MATERIALS P2, P3&P5 QUESTIONS

Compiled by: R. Muza

Q4:NOV 2011 P5

- 5 (a) (i) Define *heat capacity*.
- (ii) State how the processes of melting and boiling differ from evaporation.
- (iii) Explain the
1. cooling effect associated with evaporation,
 2. reason why water is used as a coolant in engines. [5]
- (b) Give the first law of thermodynamics and the principle upon which it is based. [2]
- (c) (i) Determine the amount of heat needed to convert
1. 5 kg of ice to water at 0°C
 2. 5 kg of water to vapour at 100°C
- [Specific latent heat of fusion of water = 336 kJ/kg
Specific latent heat of vaporisation of water = 2.0×10^6 J/kg]
- (ii) Comment on your answers to (c)(i). [5]

Q5:JUN 2014 P5

- 3 (a) (i) Define *specific latent heat of vaporisation* (l_v).
- During an experiment to determine l_v of a liquid, the readings were recorded in **Table 1.1**.

Table 1.1

reading	voltage/v	current/A	condensed mass/kg	time/s
1	12	1.0	0.014	300
2	8	0.8	0.009	300

- (ii) Calculate the value of l_v .
- (iii) Explain why the experiment was repeated for different values of voltage and current.
- (iv) Give any **one** assumption made in your calculation in (ii).

[6]

6.4 THERMAL PROPERTIES OF MATERIALS P2, P3&P5 QUESTIONS

Compiled by: R. Muza

Q6:JUN 2010 P5

- 3 (b) (i) Define *specific latent heat of vaporisation*.
- (ii) Explain why one feels cold when wind blows on one's wet body. [5]

Q7:JUN 2011 P5

- 4 (a) Define the term
- (i) *density*,
- (ii) *specific latent heat of fusion*. [2]
- (b) With reference to the structures of solids, liquids and gases, compare their densities. [4]
- (c) A woman scientist wishes to produce a magnetic alloy by mixing 20 cm³ of copper, 30 cm³ of cobalt and 50 cm³ of iron. The scientist decides that by heating iron to 2 500 K and then adding powdered samples of copper and cobalt in the suggested amounts, she will be able to have a molten mixture at the melting point of iron.

Use data in **Table 2** to answer the questions below.

Table 2

substance	density (kgm ⁻³)	melting point (K)	specific latent heat of fusion (J/kg) × 10 ⁴	liquid or solid specific heat capacity (Jkg ⁻¹ K ⁻¹)
copper	8 930	1 360	21	385
iron	7 870	1 810	14	480
cobalt	8 900	1 770	25	420

- (i) Calculate
- the heat that would be lost by iron,
 - the total energy that would be needed by copper and cobalt if iron temperature is 28°C.
- (ii) Deduce whether the scientist's decision is achievable. [6]

Q8:NOV 2012 P5

- 5 (b) (i) Define *specific latent heat of vaporisation*.
- (ii) Describe an electrical method for the determination of the specific latent heat of vaporisation. [6]

6.4 THERMAL PROPERTIES OF MATERIALS P2, P3&P5 QUESTIONS

Compiled by: R. Muza

- (c) In an experiment to determine the specific latent heat of vaporisation of a liquid, a 60 W heater is used to boil a liquid for 8 minutes and 50 g of liquid is collected.

Determine the heat lost to the surroundings given that the specific latent heat of vaporisation of the liquid is $4.3 \times 10^5 \text{ J kg}^{-1}$.

[2]

Q9: NOV 2014 P5

- 1 (d) (i) State the difference between the specific heat capacity and the specific latent heat.
- (ii) Fig. 1.3 and Fig. 1.4 show the heating curves of a liquid in a non-insulated container and an insulated container respectively. Each curve is divided into three phases I, II and III.

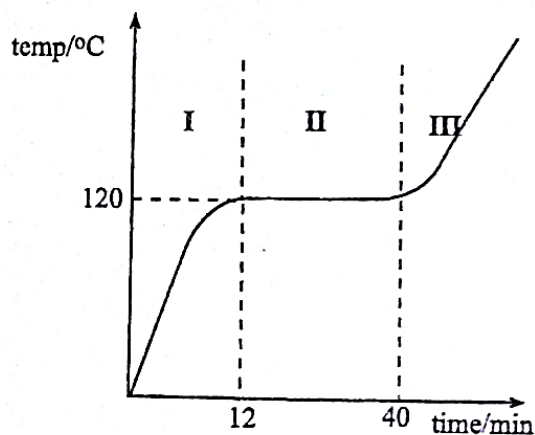


Fig. 1.3

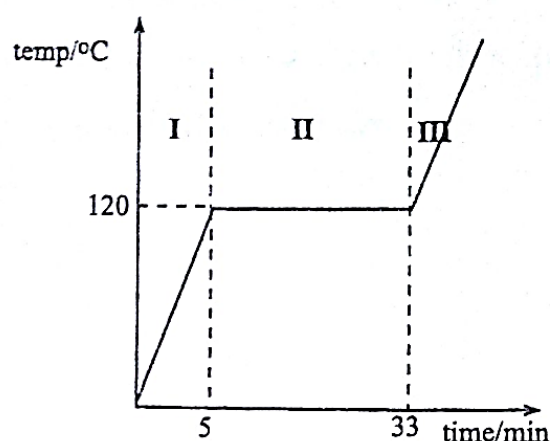


Fig. 1.4

1. Explain the differences in the shapes of the two graphs in phase I only.
2. Use Fig. 1.4 to calculate the ratio:

$$\frac{\text{specific heat capacity}}{\text{specific latent heat}}$$

[6]

Q10:NOV 2006 P3

- 6 (a) (i) Explain why the latent heat can be referred to as molecular potential energy.
- (ii) The specific latent heat of fusion of ice is 334 Jg^{-1} and the specific latent heat of vaporisation of water is $2.3 \times 10^3 \text{ Jg}^{-1}$. Explain why the specific latent heat of vaporisation is higher than the specific latent heat of fusion. [4]
- (b) Describe an experiment you would carry out to determine the specific latent heat of fusion of ice using an electrical method. [6]

Q11:NOV 2007 P3

- 6 (c) (i) Define *specific latent heat of vaporisation*.
- (ii) Explain why one feels cold when methylated spirit evaporates from the back of the hand.
- (iii) In an experiment to find the specific latent heat of vaporisation, l_v , of ethanol by an electrical method, a pupil collected 2.40 g of condensed ethanol in 30.0 s when the current flowing through the electric heater is 6.0 A and the potential difference across it is 12.0 V.
- Calculate the value of l_v of ethanol from these results.
 - The pupil feels that the value obtained is too high and decides to obtain a more accurate value by repeating the experiment using new current and potential difference values of 5.0 A and 10.0 V. He then collects 1.62 g of condensed ethanol in 30.0 s. Use both sets of results to obtain a new value of l_v .
 - Explain why this value is more accurate. [9]

Q12:JUN 2008 P3

- 6 (b) (i) Define specific latent heat of vaporization of a liquid.
- (ii) Describe and explain an electrical method for the measurement of specific latent heat of vaporization, l , of a liquid. [8]
- (c) In an experiment to determine the specific latent heat of vaporization of alcohol, 5.80 g were collected in 300 s when a current of 2.60 A and a voltage of 7.40 V were used. A mass of 11.3 g was collected in the same time when a current of 3.60 A and a voltage of 10.0 V were used.
- Calculate the
- specific latent heat of vaporization of the alcohol,
 - average rate of heat loss to the surroundings,

- (iii) power of the heater required to produce a rate of evaporation of 1.50 g per minute.

[7]

Q13:NOV 2018 P3

- 4 (b) (i) State the factors that affect the amount of heat, Q , needed to cause a certain temperature change, ΔT , of a substance.
- (ii) A man of mass 80 kg had a body temperature of 39.0 °C instead of 37.0 °C.

Assuming that a human body is 65 % water, calculate the amount of heat required to cause this temperature rise, if specific heat capacity of water is 4 200 Jkg⁻¹K⁻¹.

[6]